

Supplementary Material

Alleviating Estimation Problems in Small Sample Structural Equation Modeling—A Comparison of Constrained Maximum Likelihood, Bayesian Estimation, and Fixed Reliability Approaches

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1 Comparing Unconstrained and Constrained ML-based Estimation

Table 1.1. RMSE for parameter ab for unconstrained and constrained ML-based estimation

			Unconstrained ML		Constrained ML	
			$ML_{\lambda_1=1}$	$ML_{var(\eta)=1}$	CML_{du}	CML_{dc}
$\omega = 0.55$						
0.10	0.00	30	0.757	0.571	0.914	0.608
0.10	0.00	50	0.507	0.792	0.615	0.379
0.10	0.00	100	0.152	0.113	0.114	0.114
0.10	0.30	30	0.868	0.967	0.701	0.562
0.10	0.30	50	0.480	0.451	0.682	0.361
0.10	0.30	100	0.158	0.124	0.123	0.113
0.50	0.00	30	1.788	1.366	1.907	1.008
0.50	0.00	50	1.368	1.475	1.305	0.771
0.50	0.00	100	0.930	0.899	0.872	0.520
0.50	0.30	30	1.506	1.525	1.424	0.613
0.50	0.30	50	1.336	1.370	1.244	0.493
0.50	0.30	100	1.101	0.851	0.839	0.324
$\omega = 0.85$						
0.10	0.00	30	0.115	0.109	0.109	0.109
0.10	0.00	50	0.086	0.080	0.080	0.080
0.10	0.00	100	0.050	0.049	0.049	0.049
0.10	0.30	30	0.127	0.110	0.112	0.112
0.10	0.30	50	0.082	0.076	0.076	0.076
0.10	0.30	100	0.049	0.046	0.046	0.046
0.50	0.00	30	0.432	0.503	0.640	0.408
0.50	0.00	50	0.228	0.195	0.196	0.195
0.50	0.00	100	0.146	0.118	0.118	0.118
0.50	0.30	30	0.274	0.418	0.285	0.186
0.50	0.30	50	0.169	0.133	0.129	0.126
0.50	0.30	100	0.110	0.089	0.089	0.088

Note: In each row, the best-performing approach as well as approaches underperforming this approach by not more than 20% are marked in bold. The worst-performing approaches, defined as yielding RMSE values at least twice as high as those of the best-performing approach, are highlighted in gray. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; $ML_{\lambda_1=1}$ and $ML_{var(\eta)=1}$ refer to implementations of unconstrained ML identified by setting the first factor loading and latent variances to one; CML_{du} and CML_{dc} refer to implementations of constrained ML without and with an additional determinant constraint.

Table 1.2. RMSE for parameter c for unconstrained and constrained ML-based estimation

			Unconstrained ML		Constrained ML	
			$ML_{\lambda_1=1}$	$ML_{var(\eta)=1}$	CML_{du}	CML_{dc}
$\omega = 0.55$						
0.10	0.00	30	0.884	0.677	1.039	0.749
0.10	0.00	50	0.697	0.749	0.707	0.500
0.10	0.00	100	0.289	0.234	0.234	0.234
0.10	0.30	30	0.927	0.892	0.933	0.685
0.10	0.30	50	0.829	0.704	0.759	0.461
0.10	0.30	100	0.319	0.250	0.230	0.226
0.50	0.00	30	1.669	1.528	1.931	1.085
0.50	0.00	50	1.327	1.329	1.344	0.830
0.50	0.00	100	0.953	0.909	0.902	0.575
0.50	0.30	30	1.575	1.581	1.663	0.665
0.50	0.30	50	1.394	1.449	1.337	0.539
0.50	0.30	100	1.098	0.958	1.088	0.381
$\omega = 0.85$						
0.10	0.00	30	0.255	0.230	0.230	0.230
0.10	0.00	50	0.192	0.187	0.187	0.187
0.10	0.00	100	0.121	0.116	0.116	0.116
0.10	0.30	30	0.266	0.229	0.228	0.228
0.10	0.30	50	0.186	0.167	0.167	0.167
0.10	0.30	100	0.120	0.107	0.107	0.107
0.50	0.00	30	0.447	0.542	0.665	0.450
0.50	0.00	50	0.242	0.233	0.233	0.233
0.50	0.00	100	0.153	0.150	0.150	0.150
0.50	0.30	30	0.498	0.425	0.312	0.221
0.50	0.30	50	0.162	0.156	0.153	0.150
0.50	0.30	100	0.110	0.106	0.106	0.106

Note: In each row, the best-performing approach as well as approaches underperforming this approach by not more than 20% are marked in bold. The worst-performing approaches, defined as yielding RMSE values at least twice as high as those of the best-performing approach, are highlighted in gray. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; $ML_{\lambda_1=1}$ and $ML_{var(\eta)=1}$ refer to implementations of unconstrained ML identified by setting the first factor loading and latent variances to one; CML_{du} and CML_{dc} refer to implementations of constrained ML without and with an additional determinant constraint.

2 Comparing the Quality of Parameter Estimates and Inference of Competing Solutions for Small Sample SEM

2.1 Quality of parameter estimates

Table 2.1. Bias for parameter ab for competing solutions for small sample SEM

SEM					Fixed Reliability Single Indicator							
ab	c	N	CML_{dc}	BE_{unif}	SI_{true}	SI_5	SI_6	SI_7	SI_8	SI_9	SI_{10}	
$\omega = 0.55$												
0.10	0.00	30	-0.017	-0.067	0.020	0.048	0.002	-0.034	-0.052	-0.063	-0.070	
0.10	0.00	50	-0.011	-0.043	0.017	0.046	-0.006	-0.035	-0.051	-0.062	-0.070	
0.10	0.00	100	0.005	-0.009	0.008	0.037	-0.012	-0.037	-0.053	-0.063	-0.070	
0.10	0.30	30	-0.017	-0.055	0.002	0.032	-0.004	-0.026	-0.042	-0.053	-0.061	
0.10	0.30	50	-0.016	-0.037	0.000	0.007	-0.013	-0.034	-0.048	-0.058	-0.065	
0.10	0.30	100	-0.004	-0.017	0.003	0.025	-0.013	-0.035	-0.049	-0.059	-0.066	
0.50	0.00	30	-0.105	-0.334	0.018	0.101	-0.048	-0.171	-0.276	-0.332	-0.366	
0.50	0.00	50	0.006	-0.240	0.100	0.205	-0.014	-0.212	-0.293	-0.340	-0.371	
0.50	0.00	100	0.090	-0.107	0.086	0.263	-0.055	-0.220	-0.295	-0.340	-0.371	
0.50	0.30	30	-0.184	-0.316	-0.083	-0.063	-0.116	-0.201	-0.265	-0.313	-0.347	
0.50	0.30	50	-0.098	-0.224	-0.058	-0.011	-0.107	-0.208	-0.269	-0.315	-0.348	
0.50	0.30	100	-0.014	-0.116	0.012	0.064	-0.066	-0.188	-0.262	-0.310	-0.344	
$\omega = 0.85$												
0.10	0.00	30	-0.012	-0.043	-0.010	0.237	0.148	0.048	0.004	-0.021	-0.038	
0.10	0.00	50	-0.002	-0.027	0.002	0.309	0.147	0.061	0.017	-0.010	-0.029	
0.10	0.00	100	-0.006	-0.022	-0.006	0.267	0.112	0.045	0.007	-0.017	-0.033	
0.10	0.30	30	-0.003	-0.037	-0.003	0.141	0.079	0.043	0.008	-0.013	-0.027	
0.10	0.30	50	-0.000	-0.027	0.002	0.152	0.088	0.043	0.013	-0.008	-0.023	
0.10	0.30	100	-0.002	-0.020	-0.002	0.134	0.076	0.037	0.009	-0.011	-0.026	
0.50	0.00	30	0.032	-0.158	0.054	0.541	0.568	0.476	0.206	-0.054	-0.172	
0.50	0.00	50	0.009	-0.093	0.028	0.700	0.766	0.572	0.178	-0.065	-0.174	
0.50	0.00	100	0.001	-0.048	0.008	0.747	0.869	0.611	0.134	-0.073	-0.175	
0.50	0.30	30	-0.011	-0.137	-0.003	0.116	0.110	0.091	0.041	-0.056	-0.150	
0.50	0.30	50	0.001	-0.086	0.005	0.144	0.143	0.120	0.060	-0.055	-0.147	
0.50	0.30	100	0.002	-0.047	0.000	0.129	0.137	0.119	0.059	-0.062	-0.151	

Note: ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; CML_{dc} : constrained maximum likelihood estimation with an additional determinant constraint; BE_{unif} : Bayesian marginal estimation approach with uniform prior on standardized factor loadings and correlations; SI : fixed reliability single indicator approaches with reliabilities fixed to values stemming from the sequence $\{0.5 + 0.1i\}_{i=0}^5$ as well as to the true reliability.

Table 2.2. Bias for parameter c for competing solutions for small sample SEM

ab	c	N	SEM		Fixed Reliability Single Indicator						
			CML _{dc}	BE _{unif}	SItrue	SI5	SI6	SI7	SI8	SI9	SI10
			$\omega = 0.55$								
0.10	0.00	30	-0.016	0.010	-0.023	-0.040	-0.013	0.010	0.018	0.022	0.024
0.10	0.00	50	-0.000	0.007	-0.019	-0.037	-0.004	0.012	0.019	0.022	0.024
0.10	0.00	100	-0.015	-0.012	-0.013	-0.032	-0.001	0.012	0.018	0.021	0.023
0.10	0.30	30	-0.035	-0.158	-0.009	-0.007	-0.032	-0.061	-0.083	-0.103	-0.119
0.10	0.30	50	-0.017	-0.069	-0.009	0.022	-0.028	-0.058	-0.083	-0.103	-0.119
0.10	0.30	100	-0.021	-0.036	-0.012	0.006	-0.028	-0.058	-0.082	-0.102	-0.119
0.50	0.00	30	0.044	0.108	-0.013	-0.058	0.016	0.075	0.130	0.147	0.149
0.50	0.00	50	-0.047	0.104	-0.100	-0.162	-0.026	0.107	0.138	0.147	0.147
0.50	0.00	100	-0.112	0.033	-0.097	-0.226	0.003	0.105	0.131	0.139	0.140
0.50	0.30	30	0.035	-0.061	0.032	0.051	0.023	0.026	0.016	0.003	-0.012
0.50	0.30	50	0.032	0.021	0.043	0.040	0.044	0.051	0.033	0.016	-0.002
0.50	0.30	100	-0.016	-0.002	-0.032	-0.031	-0.010	0.012	0.008	-0.004	-0.019
$\omega = 0.85$											
0.10	0.00	30	0.005	0.010	0.003	-0.176	-0.117	-0.035	-0.005	0.009	0.016
0.10	0.00	50	-0.008	-0.005	-0.011	-0.253	-0.118	-0.051	-0.020	-0.004	0.006
0.10	0.00	100	0.011	0.011	0.009	-0.190	-0.065	-0.019	0.003	0.015	0.021
0.10	0.30	30	-0.010	-0.050	-0.002	0.090	0.072	0.036	0.011	-0.015	-0.037
0.10	0.30	50	-0.004	-0.032	-0.001	0.111	0.078	0.044	0.013	-0.014	-0.036
0.10	0.30	100	0.010	-0.009	0.013	0.159	0.107	0.063	0.028	-0.001	-0.024
0.50	0.00	30	-0.031	0.036	-0.050	-0.284	-0.387	-0.372	-0.171	0.030	-0.101
0.50	0.00	50	-0.019	-0.005	-0.036	-0.434	-0.586	-0.477	-0.155	0.030	0.092
0.50	0.00	100	0.001	-0.009	-0.002	-0.447	-0.663	-0.498	-0.096	0.051	0.106
0.50	0.30	30	-0.001	0.000	-0.010	0.061	0.029	-0.004	-0.017	0.004	0.025
0.50	0.30	50	-0.015	-0.012	-0.017	0.047	0.010	-0.024	-0.033	0.002	0.020
0.50	0.30	100	0.001	0.008	0.000	0.065	0.027	-0.009	-0.018	0.019	0.032

Note: ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; CML_{dc}: constrained maximum likelihood estimation with an additional determinant constraint; BE_{unif}: Bayesian marginal estimation approach with uniform prior on standardized factor loadings and correlations; SI: fixed reliability single indicator approaches with reliabilities fixed to values stemming from the sequence $\{0.5 + 0.1i\}_{i=0}^5$ as well as to the true reliability.

2.2 Quality of inference

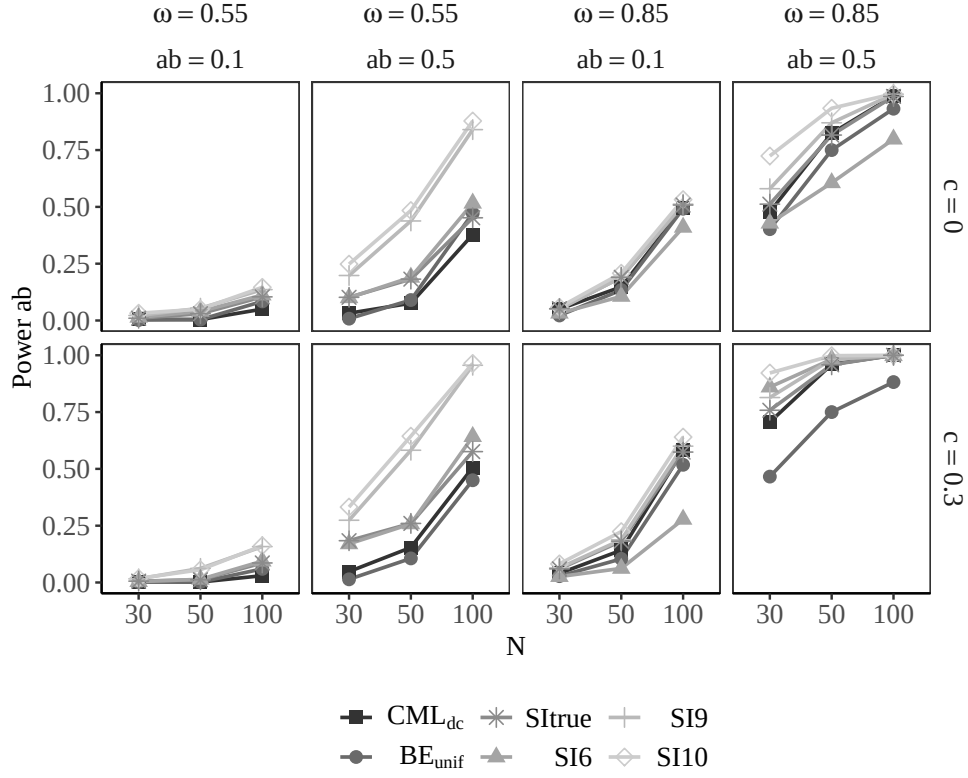


Figure 2.1. Power for parameter ab . ab gives the size of the indirect effect; N : sample size; ω : scale reliabilities; CML_{dc} : constrained maximum likelihood estimation with an additional determinant constraint; BE_{unif} : Bayesian marginal estimation approach with uniform prior on standardized factor loadings and correlations; SI : fixed reliability single indicator approaches with reliabilities fixed the true reliability, 0.60, 0.90 as well as to 1 (corresponding to a manifest approach).

3 Evaluating Bayesian Structural Equation Modeling with Weakly Informative Priors

3.1 Quality of parameter estimates

Table 3.1. RMSE for parameter c for the Bayesian approach with different priors on correlations

ab	c	N	Best	unif	$\rho_0 = 0$			$\rho_0 = \text{true}/2$			$\rho_0 = \text{true}$			
					v ₀ = 1	v ₀ = 3	v ₀ = 10	v ₀ = 1	v ₀ = 3	v ₀ = 10	v ₀ = 1	v ₀ = 3	v ₀ = 10	
$\omega = 0.55$														
0.10	0.00	30	BE _{unif}	0.182	0.182	0.167	0.145	0.097	0.174	0.154	0.109	0.183	0.164	0.122
0.10	0.00	50	SI10	0.152	0.225	0.212	0.189	0.136	0.218	0.197	0.147	0.224	0.204	0.157
0.10	0.00	100	SI10	0.110	0.205	0.195	0.180	0.144	0.197	0.183	0.149	0.201	0.188	0.156
0.10	0.30	30	SI10	0.219	0.256	0.252	0.247	0.244	0.212	0.193	0.156	0.201	0.182	0.132
0.10	0.30	50	SI10	0.181	0.208	0.199	0.192	0.192	0.191	0.173	0.142	0.193	0.174	0.131
0.10	0.30	100	SI9	0.156	0.183	0.176	0.167	0.159	0.178	0.165	0.141	0.183	0.169	0.138
0.50	0.00	30	BE _{unif}	0.236	0.236	0.220	0.191	0.136	0.265	0.246	0.198	0.283	0.261	0.205
0.50	0.00	50	SI10	0.207	0.259	0.242	0.216	0.163	0.264	0.245	0.200	0.272	0.255	0.212
0.50	0.00	100	SI10	0.174	0.261	0.247	0.225	0.182	0.261	0.243	0.201	0.274	0.264	0.235
0.50	0.30	30	SI10	0.171	0.209	0.206	0.198	0.196	0.196	0.180	0.135	0.170	0.161	0.127
0.50	0.30	50	SI10	0.132	0.203	0.191	0.173	0.143	0.201	0.185	0.143	0.185	0.174	0.142
0.50	0.30	100	SI10	0.101	0.207	0.195	0.175	0.136	0.205	0.191	0.152	0.200	0.194	0.172
$\omega = 0.85$														
0.10	0.00	30	BE _{unif}	0.187	0.187	0.178	0.164	0.131	0.180	0.166	0.134	0.181	0.168	0.139
0.10	0.00	50	SI10	0.159	0.170	0.165	0.157	0.134	0.166	0.158	0.136	0.167	0.159	0.138
0.10	0.00	100	SI10	0.102	0.111	0.110	0.107	0.099	0.110	0.107	0.100	0.110	0.108	0.100
0.10	0.30	30	SI10	0.193	0.207	0.200	0.192	0.185	0.198	0.185	0.159	0.200	0.184	0.150
0.10	0.30	50	SI10	0.145	0.160	0.156	0.152	0.149	0.155	0.148	0.134	0.157	0.149	0.128
0.10	0.30	100	SI10	0.100	0.105	0.104	0.102	0.101	0.104	0.102	0.095	0.105	0.102	0.094
0.50	0.00	30	SI10	0.227	0.253	0.238	0.215	0.169	0.253	0.233	0.191	0.272	0.259	0.228
0.50	0.00	50	SI10	0.171	0.209	0.196	0.178	0.146	0.212	0.195	0.162	0.234	0.224	0.200
0.50	0.00	100	SI9	0.147	0.161	0.153	0.143	0.124	0.163	0.153	0.134	0.179	0.171	0.157
0.50	0.30	30	SI10	0.138	0.181	0.171	0.159	0.138	0.181	0.168	0.139	0.185	0.181	0.155
0.50	0.30	50	SI10	0.106	0.154	0.146	0.137	0.124	0.156	0.144	0.125	0.174	0.169	0.146
0.50	0.30	100	SI10	0.080	0.126	0.118	0.108	0.095	0.129	0.118	0.101	0.157	0.146	0.125

Note: In each row, RMSE values outperforming those retrieved for BE_{unif} by at least 10% are printed in bold. RMSE values exceeding those retrieved for BE_{unif} by more than 10% are highlighted in gray. Best gives the best-performing approaches and RMSE values from Table 4 in the main body of the article. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif}: Bayesian approach with uniform prior with $\text{unif}(-1 + \varepsilon, 1 - \varepsilon)$ on correlations; v_0 and ρ_0 give the prior sample size and prior guess for the informative four-parameter Beta prior employed for correlations; true and true/2 correspond to the true and $0.5 \times$ the true parameter.

Table 3.2. Bias for parameter ab for the Bayesian approach with different priors on correlations

ab	c	N	unif	$\rho_0 = 0$						$\rho_0 = \text{true}/2$						$\rho_0 = \text{true}$			
				$\omega = 0.55$															
				$V_0 = 1$	$V_0 = 3$	$V_0 = 10$	$V_0 = 1$	$V_0 = 3$	$V_0 = 10$	$V_0 = 1$	$V_0 = 3$	$V_0 = 10$	$V_0 = 1$	$V_0 = 3$	$V_0 = 10$				
0.10	0.00	30	-0.067	-0.073	-0.079	-0.090	-0.049	-0.052	-0.059	-0.018	-0.013	-0.005							
0.10	0.00	50	-0.043	-0.051	-0.060	-0.078	-0.032	-0.038	-0.050	-0.010	-0.009	-0.007							
0.10	0.00	100	-0.009	-0.017	-0.029	-0.053	-0.005	-0.013	-0.030	0.008	0.005	0.001							
0.10	0.30	30	-0.055	-0.065	-0.075	-0.089	-0.035	-0.044	-0.055	0.004	0.001	-0.001							
0.10	0.30	50	-0.037	-0.048	-0.060	-0.079	-0.027	-0.036	-0.049	-0.001	-0.005	-0.006							
0.10	0.30	100	-0.017	-0.024	-0.036	-0.058	-0.012	-0.020	-0.034	0.001	-0.001	-0.004							
0.50	0.00	30	-0.334	-0.364	-0.401	-0.453	-0.260	-0.282	-0.320	-0.109	-0.104	-0.071							
0.50	0.00	50	-0.240	-0.279	-0.326	-0.407	-0.191	-0.222	-0.279	-0.068	-0.069	-0.050							
0.50	0.00	100	-0.107	-0.146	-0.204	-0.311	-0.078	-0.117	-0.195	0.016	0.009	0.006							
0.50	0.30	30	-0.316	-0.353	-0.395	-0.451	-0.232	-0.261	-0.305	0.036	0.009	0.009							
0.50	0.30	50	-0.224	-0.265	-0.315	-0.397	-0.173	-0.209	-0.266	0.042	0.020	0.012							
0.50	0.30	100	-0.116	-0.153	-0.207	-0.308	-0.084	-0.122	-0.193	0.078	0.060	0.040							
$\omega = 0.85$																			
0.10	0.00	30	-0.043	-0.048	-0.055	-0.071	-0.038	-0.042	-0.050	-0.028	-0.027	-0.023							
0.10	0.00	50	-0.027	-0.031	-0.038	-0.054	-0.024	-0.028	-0.036	-0.017	-0.017	-0.016							
0.10	0.00	100	-0.022	-0.024	-0.028	-0.039	-0.021	-0.022	-0.028	-0.017	-0.017	-0.016							
0.10	0.30	30	-0.037	-0.043	-0.052	-0.069	-0.032	-0.037	-0.046	-0.020	-0.019	-0.016							
0.10	0.30	50	-0.027	-0.031	-0.039	-0.056	-0.024	-0.027	-0.036	-0.016	-0.015	-0.013							
0.10	0.30	100	-0.020	-0.022	-0.027	-0.039	-0.018	-0.020	-0.027	-0.014	-0.013	-0.012							
0.50	0.00	30	-0.158	-0.191	-0.240	-0.335	-0.130	-0.160	-0.221	-0.047	-0.050	-0.044							
0.50	0.00	50	-0.093	-0.121	-0.165	-0.259	-0.074	-0.102	-0.163	-0.011	-0.017	-0.021							
0.50	0.00	100	-0.048	-0.065	-0.094	-0.164	-0.038	-0.056	-0.099	-0.004	-0.007	-0.012							
0.50	0.30	30	-0.137	-0.171	-0.222	-0.323	-0.107	-0.139	-0.201	0.043	0.022	0.010							
0.50	0.30	50	-0.086	-0.113	-0.154	-0.245	-0.066	-0.093	-0.149	0.041	0.027	0.013							
0.50	0.30	100	-0.047	-0.062	-0.088	-0.151	-0.036	-0.052	-0.092	0.028	0.019	0.009							

Note: ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $\text{unif}(-1 + \epsilon, 1 - \epsilon)$ on correlations; v_0 and ρ_0 give the prior sample size and prior guess for the informative four-parameter Beta prior employed for correlations; true and true/2 correspond to the true and $0.5 \times$ the true parameter.

Table 3.3. Bias for parameter c for the Bayesian approach with different priors on correlations

ab	c	N	unif	$\rho_0 = 0$			$\rho_0 = \text{true}/2$						$\rho_0 = \text{true}$			
				v ₀ = 1	v ₀ = 3	v ₀ = 10	$\omega = 0.55$			v ₀ = 10	v ₀ = 3	v ₀ = 1	v ₀ = 10	v ₀ = 1	v ₀ = 3	v ₀ = 10
0.10	0.00	30	0.010	0.007	0.008	0.009	0.019	0.017	0.016	0.027	0.019	0.007	0.007	0.002	0.002	
0.10	0.00	50	0.007	0.007	0.005	0.008	0.009	0.008	0.010	0.010	0.007	0.007	0.002	0.002	0.002	
0.10	0.00	100	-0.012	-0.011	-0.008	-0.002	-0.013	-0.011	-0.005	-0.014	-0.014	-0.014	-0.011	-0.011	-0.011	
0.10	0.30	30	-0.158	-0.169	-0.186	-0.217	-0.087	-0.087	-0.096	-0.023	-0.015	-0.007	-0.007	-0.007	-0.007	
0.10	0.30	50	-0.069	-0.082	-0.104	-0.151	-0.038	-0.049	-0.071	-0.001	-0.001	-0.001	-0.004	-0.004	-0.004	
0.10	0.30	100	-0.036	-0.044	-0.060	-0.097	-0.029	-0.038	-0.057	-0.012	-0.015	-0.015	-0.016	-0.016	-0.016	
0.50	0.00	30	0.108	0.098	0.087	0.072	0.144	0.134	0.124	0.162	0.139	0.087	0.087	0.087	0.087	
0.50	0.00	50	0.104	0.100	0.094	0.089	0.110	0.105	0.106	0.111	0.093	0.056	0.056	0.056	0.056	
0.50	0.00	100	0.033	0.040	0.052	0.075	0.025	0.032	0.056	0.003	-0.003	-0.015	-0.015	-0.015	-0.015	
0.50	0.30	30	-0.061	-0.081	-0.102	-0.151	0.015	0.008	-0.004	-0.014	-0.015	-0.031	-0.031	-0.031	-0.031	
0.50	0.30	50	0.021	0.007	-0.009	-0.056	0.037	0.030	0.016	-0.005	-0.013	-0.027	-0.027	-0.027	-0.027	
0.50	0.30	100	-0.002	-0.003	-0.006	-0.022	-0.004	-0.004	-0.002	-0.049	-0.056	-0.056	-0.056	-0.056	-0.056	
$\omega = 0.85$																
0.10	0.00	30	0.010	0.009	0.008	0.008	0.011	0.010	0.010	0.014	0.013	0.012	0.012	0.012	0.012	
0.10	0.00	50	-0.005	-0.006	-0.006	-0.006	-0.004	-0.004	-0.003	-0.002	-0.002	-0.001	-0.001	-0.001	-0.001	
0.10	0.00	100	0.011	0.011	0.010	0.007	0.012	0.011	0.010	0.013	0.013	0.012	0.012	0.012	0.012	
0.10	0.30	30	-0.050	-0.063	-0.082	-0.123	-0.042	-0.053	-0.073	-0.020	-0.022	-0.022	-0.022	-0.022	-0.022	
0.10	0.30	50	-0.032	-0.040	-0.053	-0.086	-0.027	-0.034	-0.051	-0.014	-0.015	-0.016	-0.016	-0.016	-0.016	
0.10	0.30	100	-0.009	-0.014	-0.021	-0.043	-0.007	-0.011	-0.023	-0.001	-0.001	-0.002	-0.002	-0.002	-0.002	
0.50	0.00	30	0.036	0.035	0.036	0.044	0.037	0.036	0.044	0.036	0.031	0.021	0.021	0.021	0.021	
0.50	0.00	50	-0.005	-0.002	0.003	0.015	-0.006	-0.001	0.012	-0.014	-0.013	-0.011	-0.011	-0.011	-0.011	
0.50	0.00	100	-0.009	-0.006	-0.003	0.004	-0.010	-0.007	0.001	-0.016	-0.014	-0.011	-0.011	-0.011	-0.011	
0.50	0.30	30	0.000	-0.008	-0.020	-0.053	0.003	-0.002	-0.013	-0.030	-0.031	-0.031	-0.031	-0.031	-0.031	
0.50	0.30	50	-0.012	-0.016	-0.022	-0.039	-0.012	-0.014	-0.018	-0.045	-0.044	-0.037	-0.037	-0.037	-0.037	
0.50	0.30	100	0.008	0.005	-0.000	-0.014	0.007	0.005	-0.001	-0.020	-0.016	-0.012	-0.012	-0.012	-0.012	

Note: ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $\text{unif}(-1 + \epsilon, 1 - \epsilon)$ on correlations; v_0 and ρ_0 give the prior sample size and prior guess for the informative four-parameter Beta prior employed for correlations; true and true/2 correspond to the true and $0.5 \times$ the true parameter.

Table 3.4. RMSE for parameter ab for the Bayesian approach with different priors on standardized factor loadings

ab	c	N	Best	unif	$\omega_0 = 0.55$						$\omega_0 = 0.85$			
					$\omega = 0.55$						$\omega = 0.85$			
					$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 1$	$v_0 = 3$	$v_0 = 10$				
0.10	0.00	30	SI10	0.094	0.104	0.111	0.115	0.121	0.105	0.108	0.113			
0.10	0.00	50	SI10	0.083	0.108	0.108	0.112	0.118	0.100	0.103	0.107			
0.10	0.00	100	SI8	0.068	0.090	0.083	0.086	0.092	0.076	0.077	0.078			
0.10	0.30	30	SI10	0.087	0.099	0.101	0.104	0.108	0.098	0.100	0.103			
0.10	0.30	50	SI10	0.077	0.093	0.092	0.095	0.100	0.086	0.088	0.090			
0.10	0.30	100	SI8	0.067	0.079	0.076	0.079	0.083	0.072	0.073	0.074			
0.50	0.00	30	SI9	0.359	0.361	0.346	0.336	0.326	0.362	0.354	0.343			
0.50	0.00	50	BE _{unif}	0.287	0.287	0.289	0.279	0.266	0.309	0.299	0.286			
0.50	0.00	100	BE _{unif}	0.210	0.210	0.221	0.216	0.211	0.233	0.228	0.223			
0.50	0.30	30	SI8	0.330	0.343	0.336	0.326	0.313	0.350	0.341	0.329			
0.50	0.30	50	BE _{unif}	0.264	0.264	0.274	0.266	0.254	0.290	0.282	0.270			
0.50	0.30	100	BE _{unif}	0.188	0.188	0.203	0.198	0.190	0.212	0.207	0.202			
$\omega = 0.85$														
0.10	0.00	30	SI10	0.083	0.084	0.086	0.087	0.090	0.085	0.085	0.085			
0.10	0.00	50	SI10	0.063	0.069	0.069	0.070	0.072	0.068	0.069	0.069			
0.10	0.00	100	BE _{unif}	0.047	0.047	0.048	0.048	0.048	0.048	0.048	0.048			
0.10	0.30	30	BE _{unif}	0.084	0.084	0.082	0.084	0.090	0.080	0.081	0.082			
0.10	0.30	50	SI10	0.061	0.068	0.069	0.070	0.072	0.069	0.069	0.070			
0.10	0.30	100	SI10	0.044	0.045	0.045	0.045	0.045	0.046	0.045	0.045			
0.50	0.00	30	SI10	0.230	0.235	0.238	0.232	0.223	0.249	0.243	0.235			
0.50	0.00	50	SI9	0.193	0.193	0.193	0.190	0.191	0.196	0.191	0.187			
0.50	0.00	100	CML _{dc}	0.118	0.141	0.141	0.140	0.145	0.139	0.136	0.135			
0.50	0.30	30	CML _{dc}	0.186	0.194	0.198	0.193	0.181	0.205	0.199	0.191			
0.50	0.30	50	CML _{dc}	0.126	0.142	0.144	0.141	0.135	0.148	0.144	0.138			
0.50	0.30	100	CML _{dc}	0.088	0.106	0.104	0.105	0.103	0.105	0.103	0.101			

Note: In each row, RMSE values outperforming those retrieved for BE_{unif} by at least 10% are printed in bold. RMSE values exceeding those retrieved for BE_{unif} by more than 10% are highlighted in gray. Best gives the best-performing approaches and RMSE values from Table 4 in the main body of the article. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif}: Bayesian approach with uniform prior with unif($-1 + \epsilon, 1 - \epsilon$) on standardized factor loadings; v_0 and ω_0 give the prior sample size and prior guess on reliabilities used to construct the informative four-parameter Beta priors employed for standardized factor loadings.

Table 3.5. RMSE for parameter c for the Bayesian approach with different priors on standardized factor loadings

ab	c	N	Best	unif	$\omega_0 = 0.55$				$\omega_0 = 0.85$			
					$\omega = 0.55$				$\omega = 0.85$			
					$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 10$	$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 10$
0.10	0.00	30	BE _{unif}	0.182	0.182	0.252	0.263	0.263	0.230	0.236	0.244	0.244
0.10	0.00	50	SI10	0.152	0.225	0.249	0.259	0.259	0.222	0.229	0.235	0.235
0.10	0.00	100	SI10	0.110	0.205	0.202	0.211	0.211	0.177	0.182	0.185	0.185
0.10	0.30	30	SI10	0.219	0.256	0.254	0.261	0.261	0.250	0.255	0.260	0.260
0.10	0.30	50	SI10	0.181	0.208	0.203	0.216	0.216	0.194	0.197	0.202	0.202
0.10	0.30	100	SI9	0.156	0.183	0.179	0.189	0.189	0.171	0.172	0.171	0.171
0.50	0.00	30	BE _{unif}	0.236	0.236	0.287	0.297	0.297	0.280	0.284	0.288	0.288
0.50	0.00	50	SI10	0.207	0.259	0.274	0.285	0.285	0.262	0.266	0.269	0.269
0.50	0.00	100	SI10	0.174	0.261	0.260	0.270	0.270	0.249	0.250	0.247	0.247
0.50	0.30	30	SI10	0.171	0.209	0.214	0.219	0.219	0.214	0.219	0.228	0.228
0.50	0.30	50	SI10	0.132	0.203	0.209	0.214	0.214	0.204	0.207	0.209	0.209
0.50	0.30	100	SI10	0.101	0.207	0.201	0.203	0.203	0.199	0.198	0.195	0.195
$\omega = 0.85$												
0.10	0.00	30	BE _{unif}	0.187	0.187	0.189	0.202	0.202	0.181	0.184	0.188	0.188
0.10	0.00	50	SI10	0.159	0.170	0.168	0.176	0.176	0.165	0.168	0.172	0.172
0.10	0.00	100	SI10	0.102	0.111	0.111	0.111	0.111	0.111	0.112	0.113	0.113
0.10	0.30	30	SI10	0.193	0.207	0.205	0.217	0.217	0.202	0.203	0.206	0.206
0.10	0.30	50	SI10	0.145	0.160	0.161	0.163	0.167	0.159	0.160	0.161	0.161
0.10	0.30	100	SI10	0.100	0.105	0.107	0.109	0.109	0.107	0.107	0.107	0.107
0.50	0.00	30	SI10	0.227	0.253	0.255	0.277	0.277	0.240	0.245	0.252	0.252
0.50	0.00	50	SI10	0.171	0.209	0.206	0.215	0.235	0.195	0.198	0.204	0.204
0.50	0.00	100	SI9	0.147	0.161	0.158	0.162	0.175	0.152	0.153	0.155	0.155
0.50	0.30	30	SI10	0.138	0.181	0.176	0.179	0.182	0.169	0.171	0.174	0.174
0.50	0.30	50	SI10	0.106	0.154	0.147	0.150	0.155	0.143	0.144	0.146	0.146
0.50	0.30	100	SI10	0.080	0.126	0.121	0.122	0.126	0.117	0.118	0.119	0.119

Note: In each row, RMSE values outperforming those retrieved for BE_{unif} by at least 10% are printed in bold. RMSE values exceeding those retrieved for BE_{unif} by more than 10% are highlighted in gray. Best gives the best-performing approaches and RMSE values from Table 4 in the main body of the article. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif}: Bayesian approach with uniform prior with $\text{unif}(-1 + \varepsilon, 1 - \varepsilon)$ on standardized factor loadings; v_0 and ω_0 give the prior sample size and prior guess on reliabilities used to construct the informative four-parameter Beta priors employed for standardized factor loadings.

3.2 *Quality of inference*

Table 3.6. Coverage for parameter c for the Bayesian approach with different priors on correlations

ab	c	N	unif	$\rho_0 = 0$					$\rho_0 = \text{true}/2$					$\rho_0 = \text{true}$		
				$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 1$	$v_0 = 10$	$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 1$	$v_0 = 3$	$v_0 = 10$		
				$\omega = 0.55$					$\omega = 0.85$							
0.10	0.00	30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.998	1.000		
0.10	0.00	50	0.986	0.988	0.994	0.990	0.990	0.984	0.984	0.994	0.986	0.984	0.986	0.992	0.992	
0.10	0.00	100	0.944	0.946	0.964	0.946	0.964	0.950	0.950	0.960	0.948	0.952	0.966	0.966	0.966	
0.10	0.30	30	0.994	0.992	0.990	0.992	0.990	0.996	0.996	0.998	0.992	0.992	0.992	0.990	0.992	
0.10	0.30	50	0.994	0.994	0.990	0.994	0.990	0.994	0.994	0.994	0.994	0.992	0.994	0.996	0.996	
0.10	0.30	100	0.976	0.970	0.966	0.972	0.972	0.978	0.978	0.988	0.972	0.976	0.984	0.984	0.984	
0.50	0.00	30	0.994	0.992	0.992	0.994	0.992	0.992	0.994	0.992	0.992	0.994	0.994	0.994	0.994	
0.50	0.00	50	0.978	0.978	0.984	0.982	0.982	0.976	0.976	0.976	0.976	0.998	0.994	1.000	1.000	
0.50	0.00	100	0.964	0.966	0.958	0.972	0.972	0.968	0.968	0.970	0.974	0.976	0.976	0.980	0.980	
0.50	0.30	30	0.996	0.996	0.998	0.998	0.998	0.996	0.996	0.996	1.000	1.000	1.000	1.000	1.000	
0.50	0.30	50	0.996	0.994	0.996	0.996	0.996	0.996	0.996	0.996	1.000	1.000	1.000	1.000	1.000	
0.50	0.30	100	0.986	0.982	0.990	0.990	0.990	0.982	0.982	0.986	0.996	0.994	0.994	0.996	0.996	
$\omega = 0.85$																
0.10	0.00	30	0.966	0.970	0.972	0.984	0.966	0.972	0.986	0.962	0.970	0.984	0.962	0.970	0.984	
0.10	0.00	50	0.940	0.942	0.944	0.964	0.944	0.948	0.966	0.940	0.942	0.960	0.940	0.942	0.960	
0.10	0.00	100	0.958	0.954	0.960	0.966	0.956	0.956	0.964	0.958	0.960	0.960	0.958	0.960	0.960	
0.10	0.30	30	0.946	0.952	0.950	0.930	0.950	0.960	0.962	0.964	0.966	0.980	0.966	0.966	0.980	
0.10	0.30	50	0.940	0.944	0.944	0.940	0.944	0.948	0.950	0.946	0.952	0.964	0.946	0.952	0.964	
0.10	0.30	100	0.954	0.956	0.954	0.952	0.958	0.960	0.962	0.958	0.962	0.964	0.958	0.958	0.964	
0.50	0.00	30	0.964	0.968	0.974	0.984	0.970	0.972	0.982	0.974	0.974	0.978	0.974	0.974	0.978	
0.50	0.00	50	0.968	0.968	0.966	0.972	0.968	0.968	0.970	0.970	0.972	0.976	0.970	0.972	0.976	
0.50	0.00	100	0.962	0.960	0.958	0.966	0.960	0.960	0.964	0.958	0.960	0.962	0.958	0.960	0.962	
0.50	0.30	30	0.982	0.982	0.982	0.986	0.982	0.982	0.986	1.000	0.982	0.986	1.000	0.982	0.986	
0.50	0.30	50	0.964	0.964	0.966	0.972	0.968	0.964	0.972	0.988	0.976	0.978	0.988	0.976	0.978	
0.50	0.30	100	0.964	0.964	0.962	0.966	0.966	0.964	0.964	0.948	0.958	0.964	0.948	0.958	0.964	

Note: Coverage value between 0.925 and 0.975 are printed in bold. Severe undercoverage below 0.850 is highlighted in gray. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $\text{unif}(-1 + \varepsilon, 1 - \varepsilon)$ on correlations; v_0 and ρ_0 give the prior sample size and prior guess for the informative four-parameter Beta prior employed for correlations; true and true/2 correspond to the true and $0.5 \times$ the true parameter.

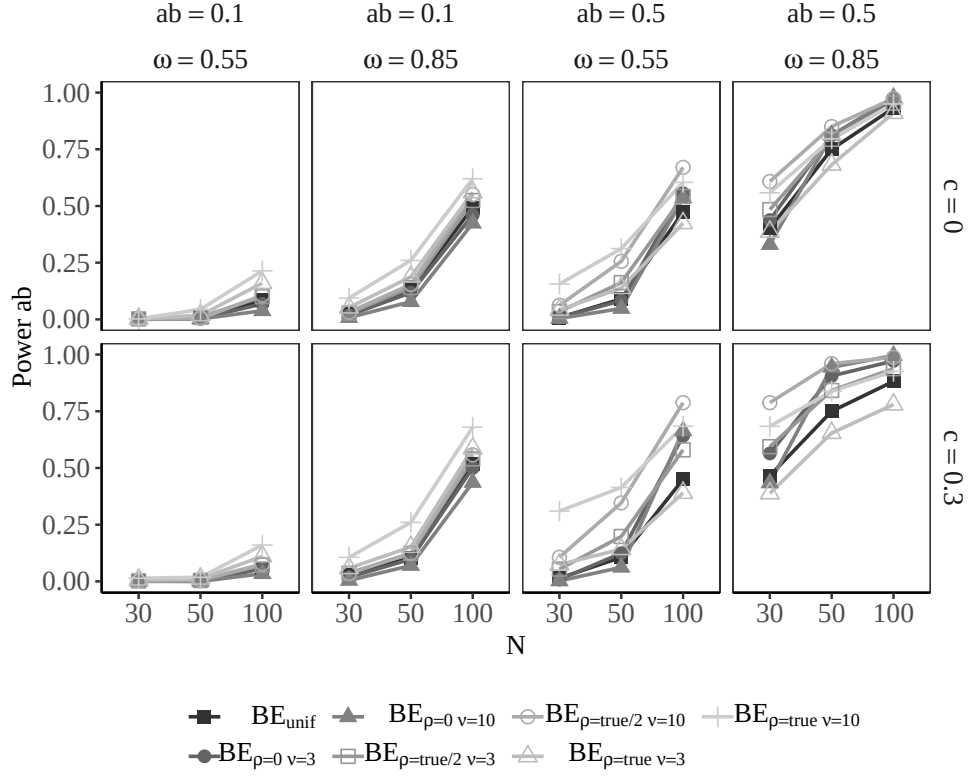


Figure 3.1. Power for parameter ab under different priors on correlations. ab gives the size of the indirect effect; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian marginal estimation approach with uniform prior on standardized factor loadings and correlations; the subscripts ρ and v give the prior guess and prior sample size for the informative four-parameter Beta prior employed for correlations; true and true/2 correspond to the true and $0.5 \times$ the true parameter.

Table 3.7. Coverage for parameter ab for the Bayesian approach with different priors on standardized factor loadings

ab	c	N	unif	$\omega_0 = 0.55$				$\omega_0 = 0.85$			
				$v_0 = 1$	$v_0 = 3$	$v_0 = 10$		$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	
				$\omega = 0.55$				$\omega = 0.85$			
0.10	0.00	30	0.998	0.998	0.998	1.000		0.992	0.996	1.000	
0.10	0.00	50	0.984	0.974	0.980	0.986		0.940	0.950	0.954	
0.10	0.00	100	0.952	0.948	0.948	0.948		0.900	0.910	0.920	
0.10	0.30	30	0.998	0.992	0.994	0.994		0.974	0.972	0.972	
0.10	0.30	50	0.974	0.962	0.966	0.968		0.920	0.932	0.940	
0.10	0.30	100	0.954	0.940	0.944	0.952		0.920	0.936	0.934	
0.50	0.00	30	0.966	0.916	0.928	0.942		0.788	0.802	0.812	
0.50	0.00	50	0.932	0.906	0.916	0.932		0.812	0.814	0.818	
0.50	0.00	100	0.964	0.928	0.932	0.950		0.886	0.894	0.890	
0.50	0.30	30	0.898	0.862	0.876	0.898		0.718	0.732	0.756	
0.50	0.30	50	0.916	0.882	0.902	0.910		0.782	0.790	0.806	
0.50	0.30	100	0.936	0.882	0.900	0.926		0.846	0.858	0.826	
$\omega = 0.85$											
0.10	0.00	30	0.918	0.932	0.940	0.948		0.910	0.926	0.940	
0.10	0.00	50	0.932	0.924	0.924	0.938		0.922	0.928	0.932	
0.10	0.00	100	0.918	0.938	0.938	0.938		0.922	0.936	0.944	
0.10	0.30	30	0.898	0.904	0.912	0.906		0.898	0.912	0.912	
0.10	0.30	50	0.916	0.908	0.928	0.926		0.920	0.926	0.930	
0.10	0.30	100	0.930	0.926	0.924	0.928		0.932	0.932	0.936	
0.50	0.00	30	0.900	0.900	0.918	0.952		0.838	0.864	0.900	
0.50	0.00	50	0.918	0.904	0.924	0.964		0.884	0.896	0.914	
0.50	0.00	100	0.914	0.922	0.930	0.952		0.900	0.910	0.918	
0.50	0.30	30	0.872	0.852	0.892	0.940		0.804	0.826	0.854	
0.50	0.30	50	0.890	0.870	0.882	0.922		0.846	0.860	0.888	
0.50	0.30	100	0.924	0.896	0.898	0.932		0.880	0.892	0.906	

Note: Coverage value between 0.925 and 0.975 are printed in bold. Severe undercoverage below 0.850 is highlighted in gray. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $unif(-1 + \varepsilon, 1 - \varepsilon)$ on standardized factor loadings; v_0 and ω_0 give the prior sample size and prior guess on reliabilities used to construct the informative four-parameter Beta priors employed for standardized factor loadings.

Table 3.8. Coverage for parameter c for the Bayesian approach with different priors on standardized factor loadings

ab	c	N	unif	$\omega_0 = 0.55$			$\omega_0 = 0.85$		
				$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 1$	$v_0 = 3$	$v_0 = 10$
0.10	0.00	30	1.000	0.988	0.990	0.988	0.980	0.984	0.974
0.10	0.00	50	0.986	0.968	0.966	0.966	0.964	0.958	0.950
0.10	0.00	100	0.944	0.938	0.938	0.934	0.936	0.932	0.936
0.10	0.30	30	0.994	0.984	0.984	0.978	0.976	0.968	0.958
0.10	0.30	50	0.994	0.982	0.988	0.990	0.978	0.980	0.972
0.10	0.30	100	0.976	0.968	0.970	0.964	0.956	0.962	0.968
0.50	0.00	30	0.994	0.988	0.986	0.988	0.968	0.968	0.974
0.50	0.00	50	0.978	0.974	0.974	0.974	0.962	0.956	0.950
0.50	0.00	100	0.964	0.962	0.958	0.960	0.954	0.950	0.940
0.50	0.30	30	0.996	0.994	0.994	0.994	0.988	0.986	0.984
0.50	0.30	50	0.996	0.984	0.988	0.992	0.978	0.978	0.980
0.50	0.30	100	0.986	0.980	0.982	0.984	0.970	0.970	0.970
$\omega = 0.85$									
0.10	0.00	30	0.966	0.966	0.962	0.966	0.966	0.964	0.966
0.10	0.00	50	0.940	0.942	0.944	0.944	0.942	0.948	0.946
0.10	0.00	100	0.958	0.958	0.954	0.954	0.954	0.954	0.952
0.10	0.30	30	0.946	0.958	0.954	0.958	0.946	0.958	0.956
0.10	0.30	50	0.940	0.940	0.936	0.936	0.940	0.940	0.940
0.10	0.30	100	0.954	0.952	0.948	0.952	0.954	0.956	0.954
0.50	0.00	30	0.964	0.974	0.974	0.964	0.954	0.958	0.968
0.50	0.00	50	0.968	0.964	0.970	0.970	0.960	0.968	0.966
0.50	0.00	100	0.962	0.958	0.960	0.960	0.958	0.964	0.958
0.50	0.30	30	0.982	0.978	0.978	0.986	0.982	0.978	0.976
0.50	0.30	50	0.964	0.970	0.970	0.980	0.972	0.970	0.968
0.50	0.30	100	0.964	0.966	0.966	0.974	0.968	0.968	0.970

Note: Coverage value between 0.925 and 0.975 are printed in bold. Severe undercoverage below 0.850 is highlighted in gray. ab and c give the size of the indirect and direct effect, respectively; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $\text{unif}(-1 + \epsilon, 1 - \epsilon)$ on standardized factor loadings; v_0 and ω_0 give the prior sample size and prior guess on reliabilities used to construct the informative four-parameter Beta priors employed for standardized factor loadings.

Table 3.9. Type I error for parameter c under $c = 0.00$ for the Bayesian approach with different priors on standardized factor loadings

ab	N	unif	$\omega_0 = 0.55$						$\omega_0 = 0.85$								
			$\omega = 0.55$						$\omega = 0.85$								
			$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 1$	$v_0 = 3$	$v_0 = 10$	$v_0 = 1$	$v_0 = 3$	$v_0 = 10$						
0.10	30	0.000	0.012	0.010	0.012	0.020	0.016	0.026	0.10	30	0.034	0.038	0.034	0.034	0.036	0.034	
0.10	50	0.014	0.032	0.034	0.034	0.034	0.036	0.042	0.050	0.10	50	0.060	0.056	0.056	0.052	0.052	0.054
0.10	100	0.056	0.062	0.062	0.066	0.066	0.064	0.068	0.064	0.10	100	0.042	0.046	0.046	0.046	0.046	0.048
0.50	30	0.006	0.012	0.014	0.012	0.012	0.032	0.032	0.026	0.50	30	0.036	0.026	0.036	0.046	0.042	0.032
0.50	50	0.022	0.026	0.026	0.026	0.026	0.038	0.044	0.050	0.50	50	0.032	0.030	0.030	0.032	0.032	0.034
0.50	100	0.036	0.038	0.042	0.040	0.046	0.050	0.050	0.060	0.50	100	0.038	0.040	0.040	0.036	0.036	0.042

Note: Type I error rates between 0.025 and 0.075 are printed in bold. ab gives the size of the indirect effect; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $unif(-1 + \epsilon, 1 - \epsilon)$ on standardized factor loadings; v_0 and ω_0 give the prior sample size and prior guess on reliabilities used to construct the informative four-parameter Beta priors employed for standardized factor loadings.

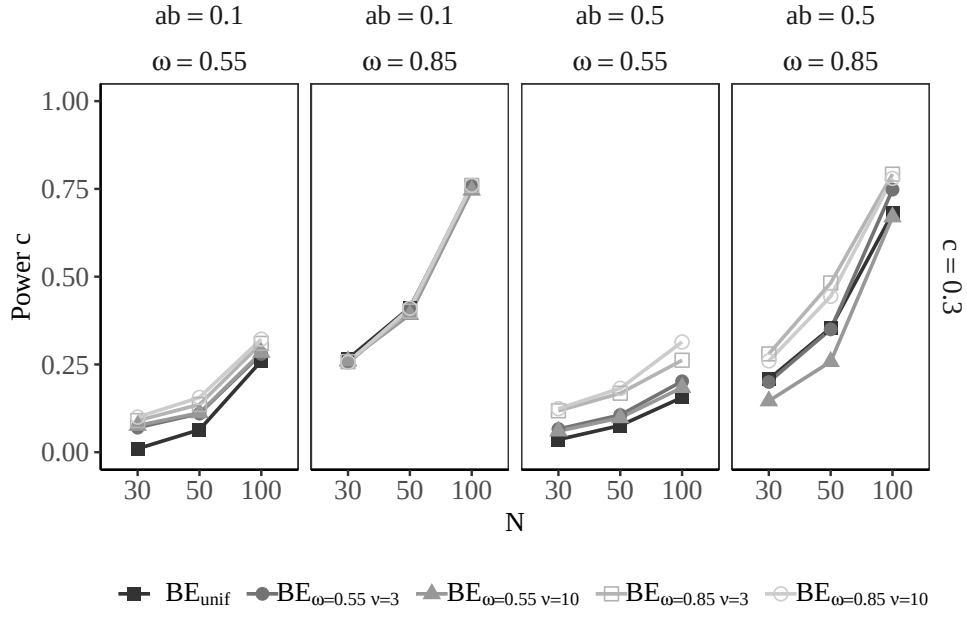


Figure 3.2. Power for parameter c under different priors on standardized factor loadings. ab gives the size of the indirect effect; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $\text{unif}(-1 + \varepsilon, 1 - \varepsilon)$ on standardized factor loadings; the subscripts ω and v give the prior guess on reliabilities and prior sample size used to construct the informative four-parameter Beta priors employed for standardized factor loadings.

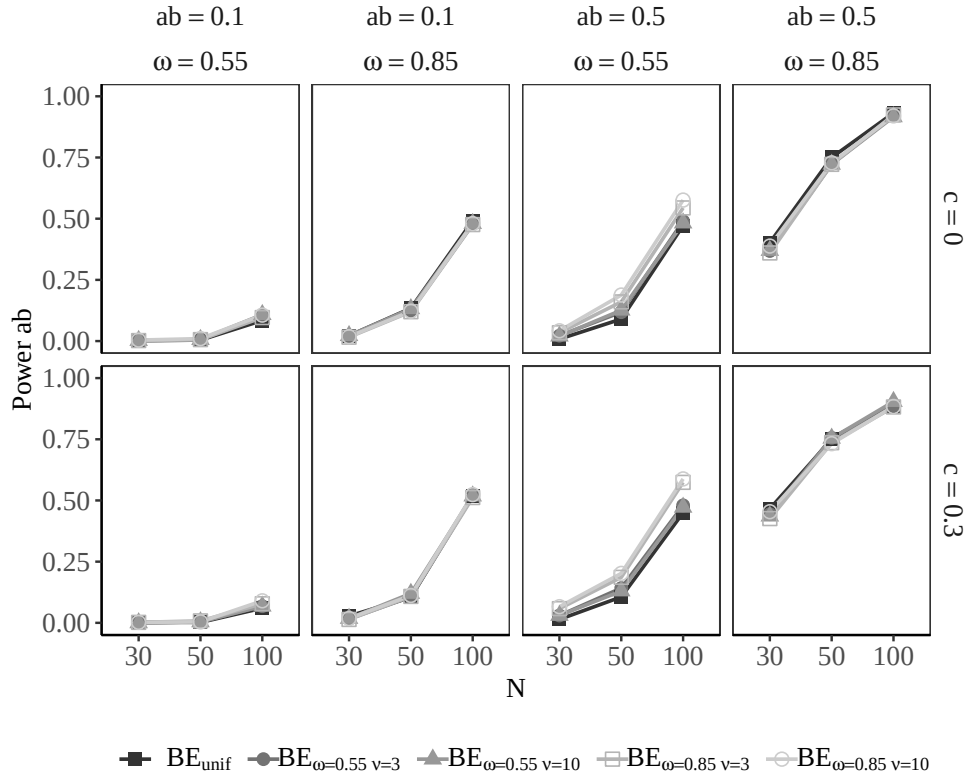


Figure 3.3. Power for parameter ab under different priors on standardized factor loadings. ab gives the size of the indirect effect; N : sample size; ω : scale reliabilities; BE_{unif} : Bayesian approach with uniform prior with $unif(-1 + \epsilon, 1 - \epsilon)$ on standardized factor loadings; the subscripts ω and v give the prior guess on reliabilities and prior sample size used to construct the informative four-parameter Beta priors employed for standardized factor loadings.